Shop Drawing Review Guide for Panel Projects

A structured guide for checking panel layout, accessories, flashing logic, quantities, and execution coordination.

GUIDE | Execution Details | 12 pages | February 2026

Purpose of Shop Drawing Review

The shop drawing is the last point where a panel project can be corrected on paper instead of on site. Reviewing it before approval is the cheapest quality step in the whole project — and skipping it is the most common cause of avoidable delay and leakage.

Why the buyer / contractor must review before approving

- Errors in shop drawings mean the wrong material is delivered.
- Missing accessories discovered late cause project delay.
- Flashing conflicts cause leakage after handover.
- An approved drawing transfers responsibility to the approver.

What a Complete Shop Drawing Package Includes

Before reviewing content, confirm the package is complete. A panel project shop-drawing set should contain every document below.

Item	Pass ✓	Fail ✗	Notes
Roof plan with panel layout and span direction.			
Wall elevations with panel setting-out.			
Panel schedule (type, length, quantity).			
Accessory / trim schedule.			
Flashing details (ridge, eave, valley, wall junction).			
Ridge / eave / valley section details.			
Penetration details.			
Fastener schedule (type, spacing).			
Quantity take-off sheet.			

Section 1: Panel Layout Check

Item	Pass ✓	Fail ✗	Notes
Panel orientation correct for the span direction.			
Panel numbering sequence is logical and matches the plan.			
Start-panel position is marked.			
Panel widths account for structural tolerance.			
Camber / slope direction shown.			
Panel lengths match the roof / wall geometry.			
Side-lap direction is away from prevailing rain.			
End-lap positions coordinated with purlins.			
Cut panels at edges are not slivers.			
Layout matches the architectural setting-out.			

Section 2: Flashing Logic Check

Item	Pass ✓	Fail ✗	Notes
Ridge flashing shown with correct overlap.			
Eave flashing with anti-capillary groove.			
Wall-to-roof junction fully detailed.			
All corners shown (hip, valley, internal).			
Flashing material and gauge specified.			
Flashing laps minimum 150 mm and sealed.			
Concealed or sealed fixings on weather-side flashings.			
Thermal movement allowance in long flashing runs.			
Parapet cap and cleat detailed.			
Flashing-to-penetration junctions shown.			

Section 3: Accessories & Trim Check

Item	Pass ✓	Fail ✗	Notes
All trim pieces listed in the schedule.			
Closure strips (foam or metal) shown at eaves and ridges.			
Gutter profile and bracket spacing shown.			
Downpipe positions and sizes shown.			
Fastener types match the substrate and skin.			
Sealant type and locations scheduled.			
Filler blocks at profile ends specified.			
Trim colors / coatings match the panels.			

Section 4: Penetrations & Openings

Item	Pass ✓	Fail ✗	Notes
All known penetrations marked on the plan.			
Rooflights / skylights coordinated with panel layout.			
HVAC curb details shown.			
Door / window head flashings detailed.			
Upstand heights specified (minimum 150 mm).			
Penetration positions avoid ridge and lap lines.			
Reinforcement / trimmer framing shown where needed.			
Sealing method specified for each penetration type.			

Section 5: Quantity Take-Off Verification

Item	Pass ✓	Fail ✗	Notes
Panel count matches the layout plan.			
Accessory quantities include an allowance for waste / cut.			
Fastener count matches the fixing schedule.			
Sealant quantities estimated for all joints.			
Flashing lengths totalled per type.			
Gutter and downpipe lengths totalled.			
Overage percentages applied per item category.			
Take-off cross-checked against the panel schedule.			

Section 6: Coordination with Other Trades

Item	Pass ✓	Fail ✗	Notes
Structural purlin layout matches the panel span.			
MEP penetration positions agreed.			
Cladding / panel interface at eaves coordinated.			
Gutter discharge to civil drainage confirmed.			
Edge / parapet interface with other trades resolved.			
Access and safety provisions coordinated for install.			

Common Shop Drawing Errors in Iranian Projects

Error	Frequency	Impact	How to Catch
Missing valley detail	Common	Leakage at valley	Check every internal corner on plan
Wrong panel orientation	Common	Re-order / rework	Verify span direction vs purlins
No thermal break shown	Frequent	Condensation, energy loss	Check wall-junction sections
Gutter undersized	Frequent	Overflow in heavy rain	Recheck catchment vs gutter size
Penetration not shown	Common	Site cutting, leak risk	Cross-check MEP drawings
Wrong accessory spec	Occasional	Delivery mismatch	Verify trim schedule vs details

Review Sign-off Matrix

Record each reviewed drawing. Status: A = approved, B = approved with comments, C = revise and resubmit.

Drawing No.	Rev	Reviewed by	Date	Status	Comments

SIPANEL Engineering Support

SIPANEL reviews shop drawings as part of its pre-purchase engineering support. Before you approve a panel package, our engineers can check layout, flashing logic, and quantities against your project geometry and climate.

This reduces the risk of wrong deliveries, late accessory discovery, and post-handover leakage — the three failures this guide is built to prevent.

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